

Focus on Function: a case study on how translating complex medical topics and explaining diagnostic uncertainty changed the course of rehabilitation for a patient with non-traumatic spinal cord injury

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Case Summary
Thirty-eight year old man with idiopathic longitudinally extensive cervicothoracic-spine transverse myelitis and optic nerve enhancement with preceding meningoencephalitis.

Presentation
Patient with no significant medical history presented to inpatient rehabilitation with quadriparesis and ataxia (limb and truncal), proprioceptive deficits, urinary retention, and orthostatic hypotension after extensive neurological workup. He was admitted with working diagnosis of CSF inflammation with long tract cervicothoracic transverse myelitis and radiographic optic nerve enhancement of unknown etiology.
When asked on admission what he understood about his condition and diagnosis, he expressed worry and frustration regarding the uncertain illness etiology and treatment plan. He demonstrated poor understanding of the changes in his body and felt hopeless about his abrupt loss of basic body function control: quadriparesis, neurogenic orthostatic hypotension, urinary retention, and nocturnal polyuria. His wife was distraught by his condition and the various medical teams’ inability to secure a unifying, parsimonious diagnosis.

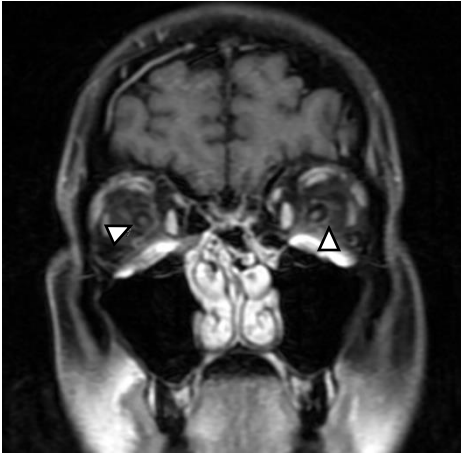


Fig 1. MRI brain showing bilateral optic nerve enhancement.

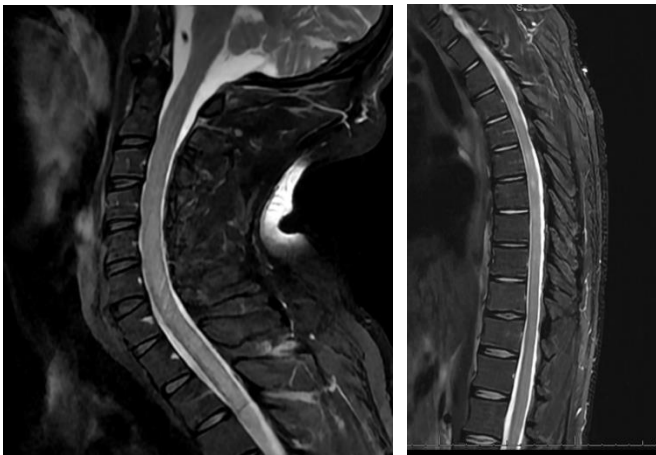


Fig 2–3. MRI demonstrating acute C-T spine transverse myelitis

Test	Result
Cytometry	↓ CD4:CD8
Autoimmune encephalitis	⊖
NMO / MOG	⊖
ANA, etc.	⊖
Metagenomic sequencing	⊖
ACE	⊖
<i>T. pallidum</i> /HIV	⊖
Anti-Hu	⊖

Fig 3. Selected Serologies. Evaluating possible underlying etiologies.

CSF results	
WBC count	373 cells/μL
Glucose	36 mg/dL
Protein	296 mg/dL
IgG	10.9 mg/dL
Albumin	82.8 mg/dL

Fig 4. Lumbar Puncture results. Suggestive of inflammatory process of unclear, noninfectious origin.

Assessment:
The differential diagnosis included neuromyelitis optica (NMO) spectrum disorder; multiple sclerosis; myelin oligodendrocyte glycoprotein antibody disease (MOG); atypical viral meningoencephalitis with transverse myelitis; neurosarcoidosis; versus paraneoplastic syndrome.

Discussion and Rehabilitation Plan
In the face of diagnostic uncertainty and abrupt functional decline, the care team collaborated on how to effectively engage this patient’s agency. The conversation began with sitting down with the patient and his wife, listening, and exploring the patient’s understanding of the complex situation. Through this level-setting conversation, empathy and trust-building were prioritized over diagnostic aims. Particular effort was devoted to explaining the anatomy, prior testing, and future workup plans, while taking care to confirm understanding and access to additional resources for review.
With a better grasp of his condition, the conversations organically shifted focus to the importance of engagement in the rehabilitation efforts. After coming to consensus with the care team on the comprehensive evaluation and treatment plan, the patient and his family participated in teach-back, further cementing their agency and partnership with the rehabilitation team. Finally, he was connected to peer mentors and support groups. Through this patient centered focus on function, he and his family were empowered to overcome barriers to progress.

Outcome and Conclusions
On admission, the patient’s mobility required minimal-to-moderate assistance at wheelchair level and ADLs required setup-to-moderate assistance; on discharge mobility improved to contact guard assist with front-wheeled walker, and his ADLs ranged from touching-assist to independent. His orthostatic hypotension and urinary impairments were addressed with medical and therapy interventions. While the neuro-immunologic and follow-up testing have yet to demonstrate an ultimate underlying etiology, the patient continues to progress in his rehabilitation journey.
His case serves as an important reminder that helping patients view their physical deficits through the lens of function — focusing on personal goals — empowers agency to accept diagnostic uncertainty, enhance their understanding of complex conditions, and improve rehabilitation outcomes.

Acknowledgements
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